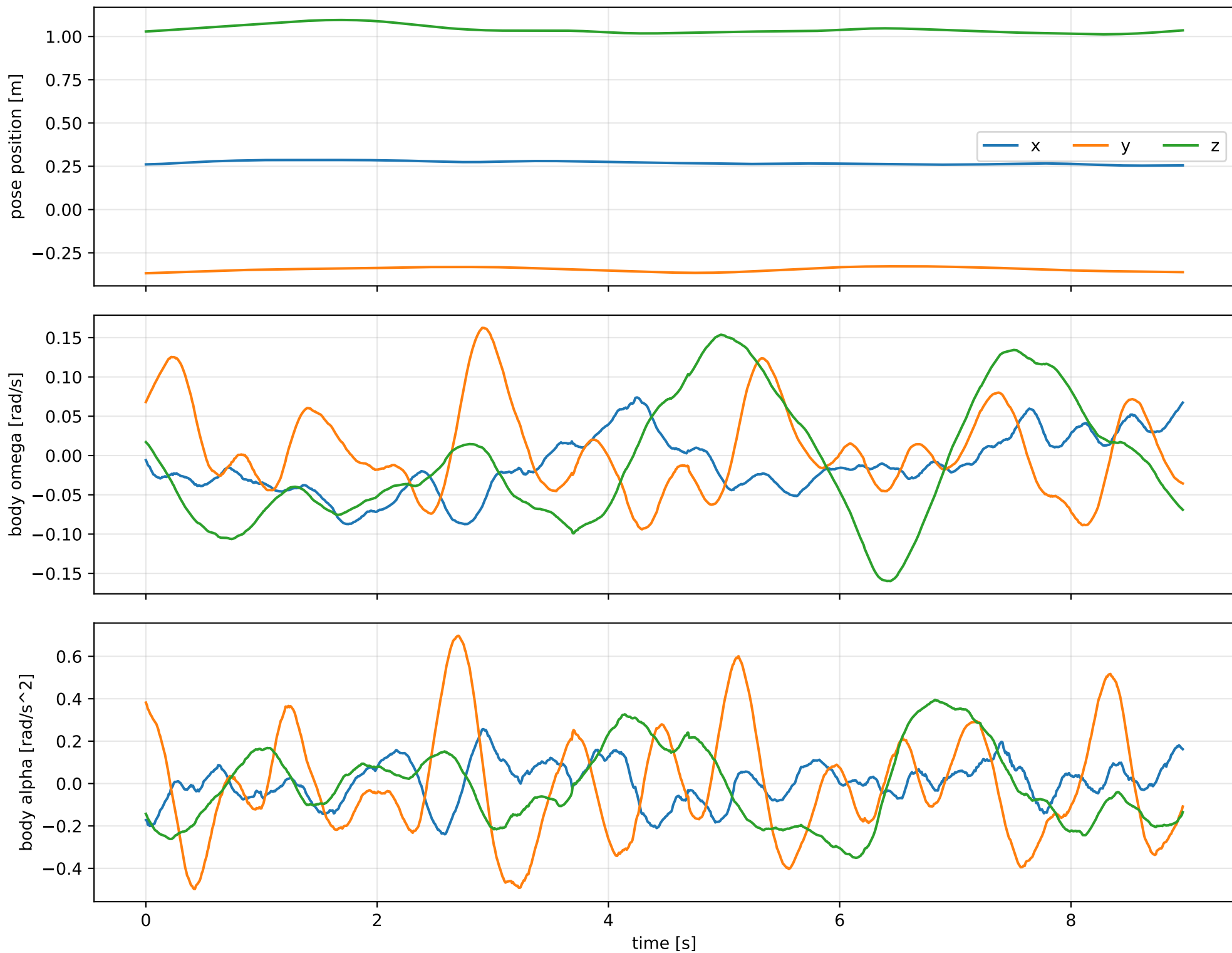
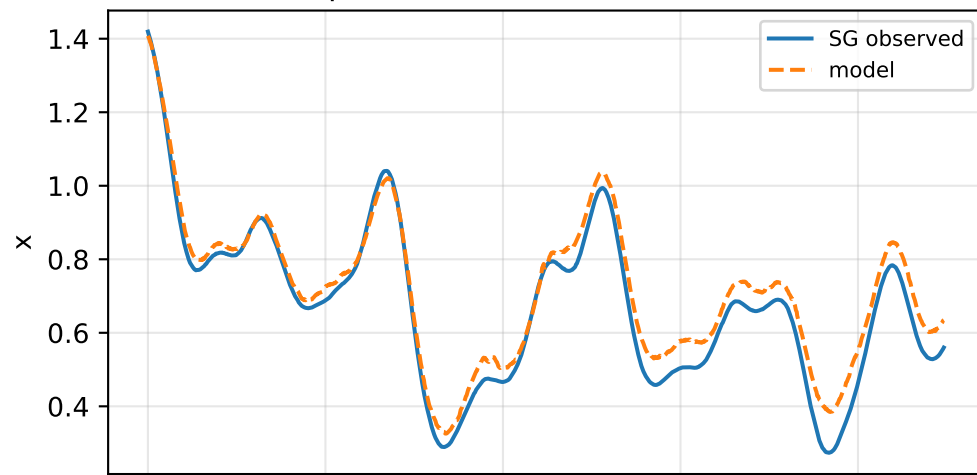


so3_naive_rotation_vector_derivatives: SG trajectory and derived motion

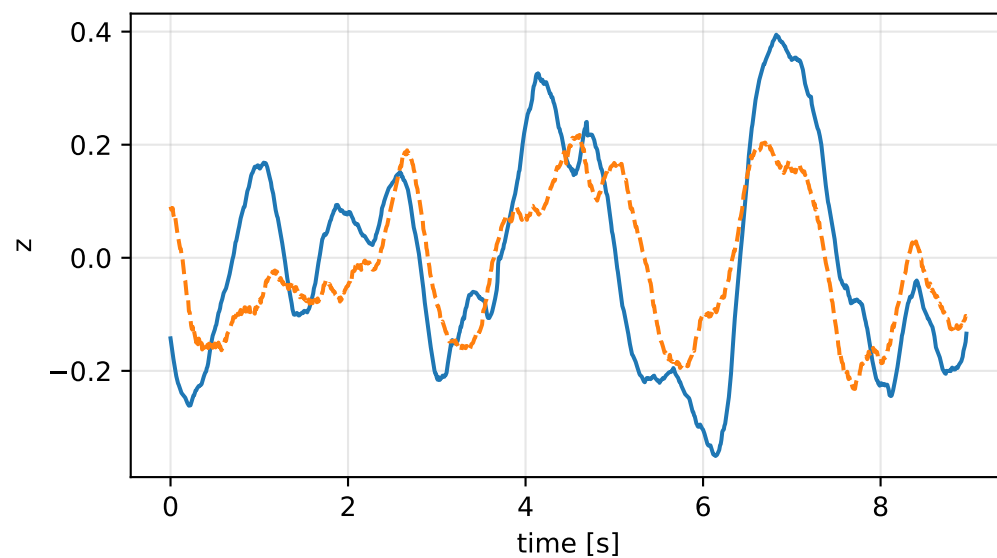
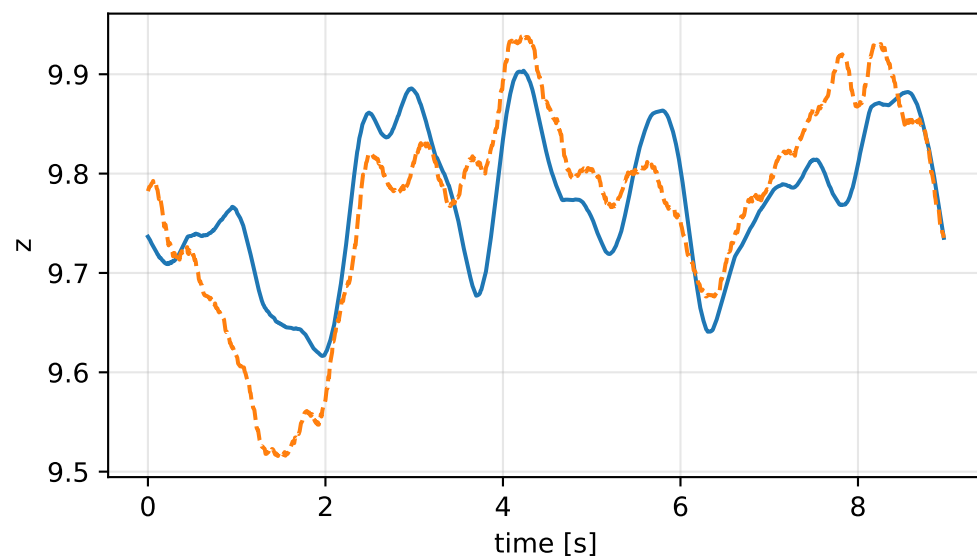
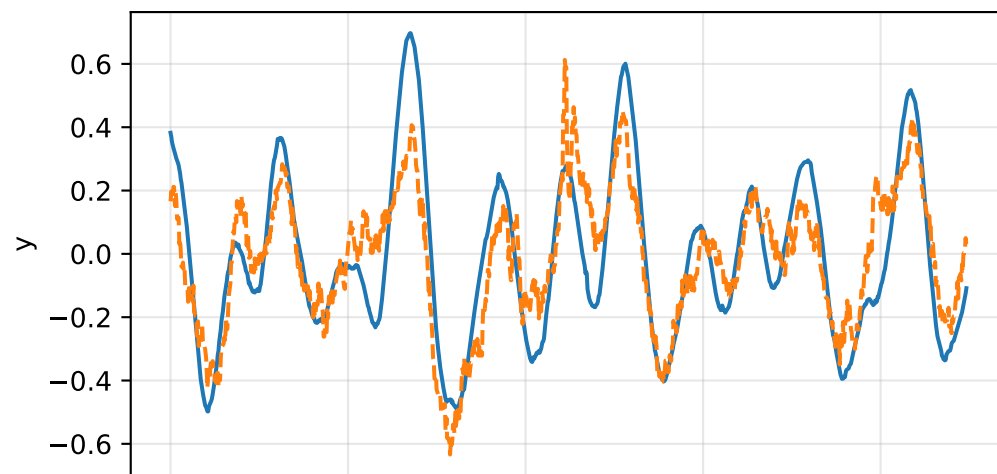
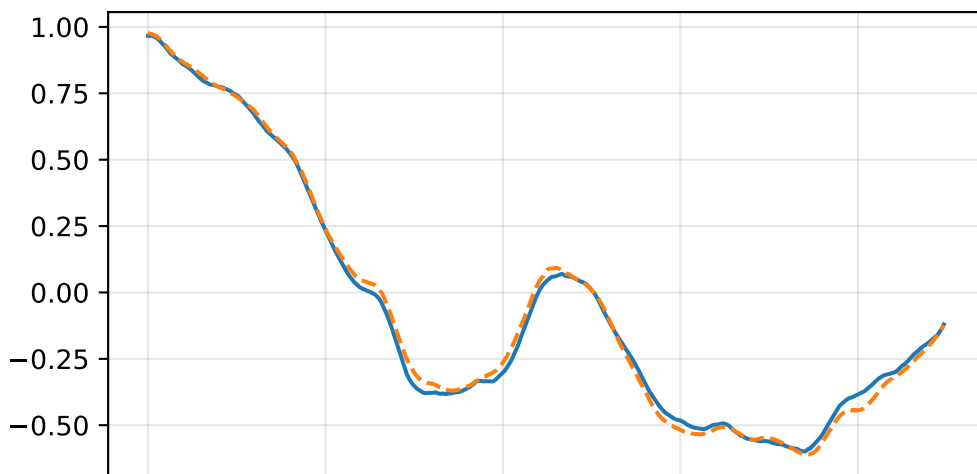
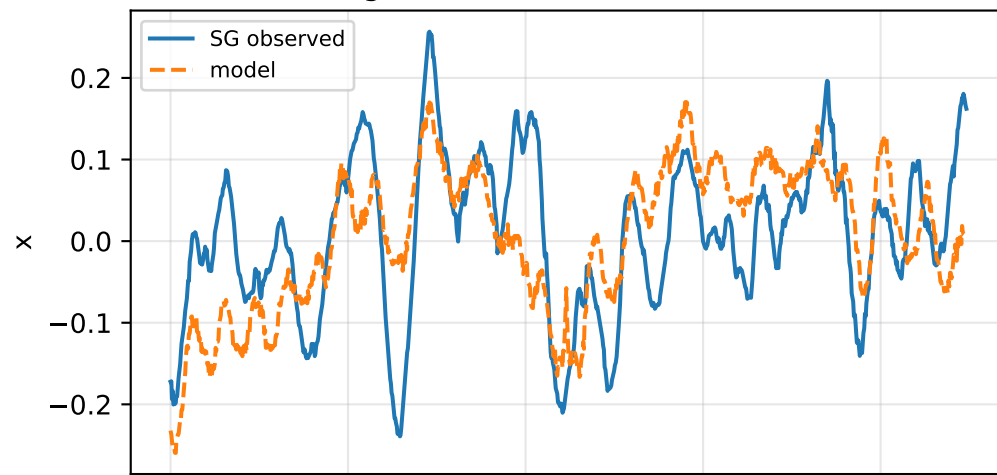


so3_naive_rotation_vector_derivatives: acceleration residual objective

specific acceleration [m/s²]

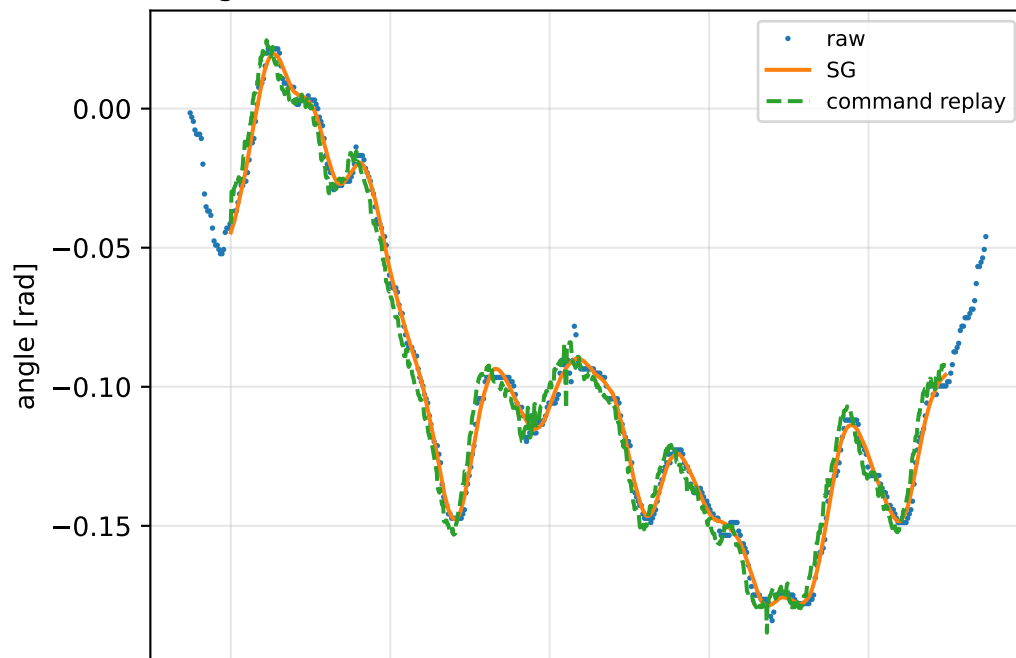


angular acceleration [rad/s²]

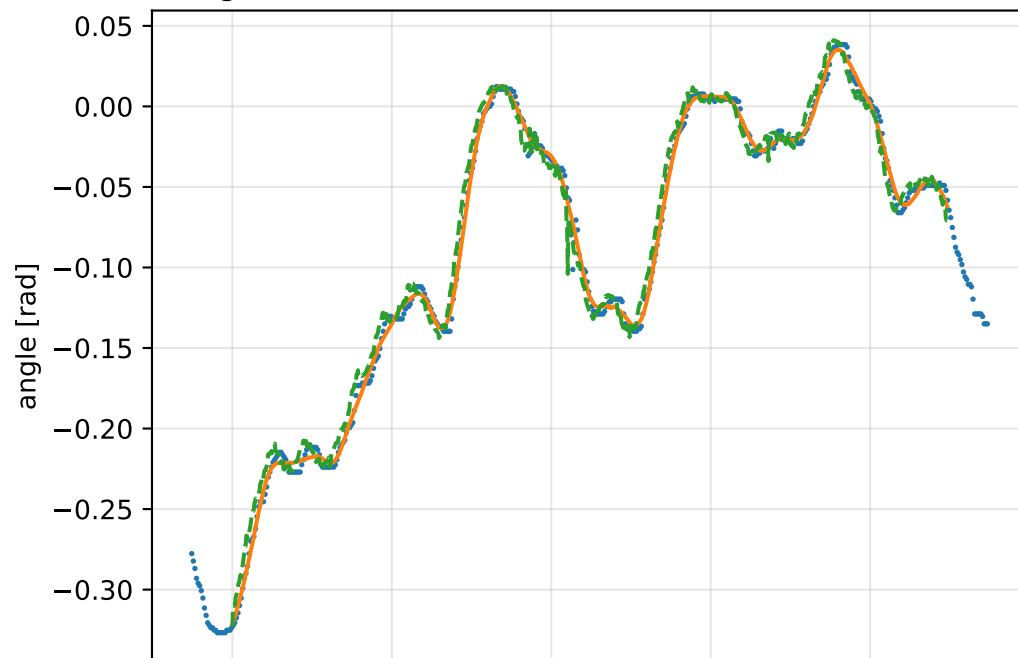


so3_naive_rotation_vector_derivatives: gimbal measurement audit (objective=measured_sg)

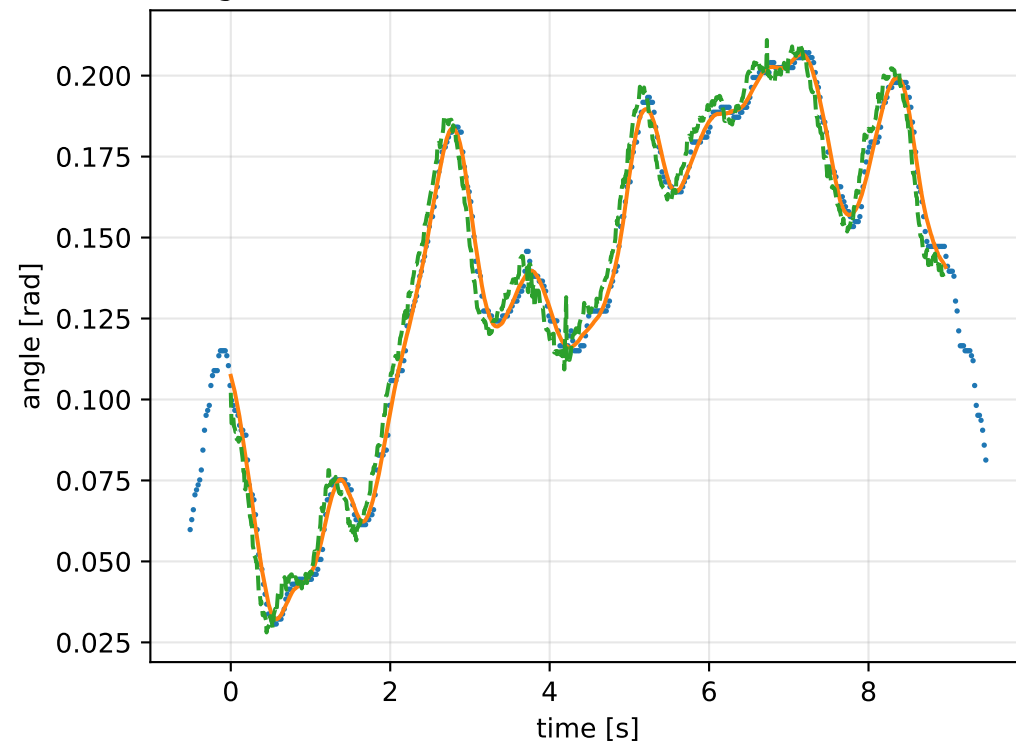
gimbal 1: RMSE=0.006345, max=0.01494 rad



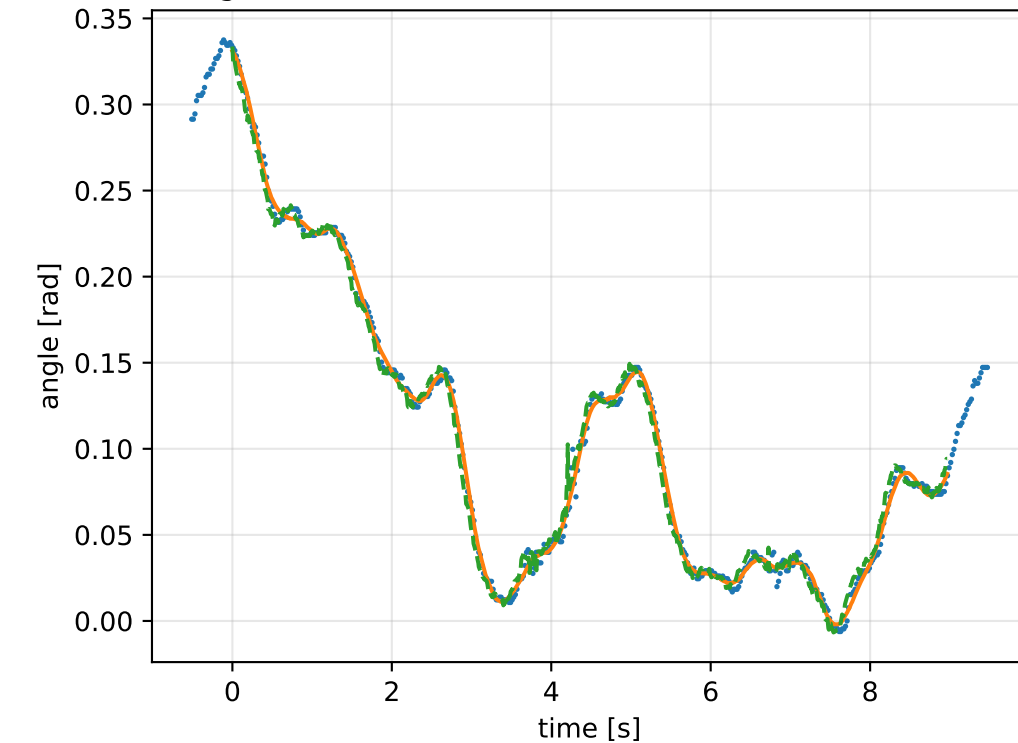
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

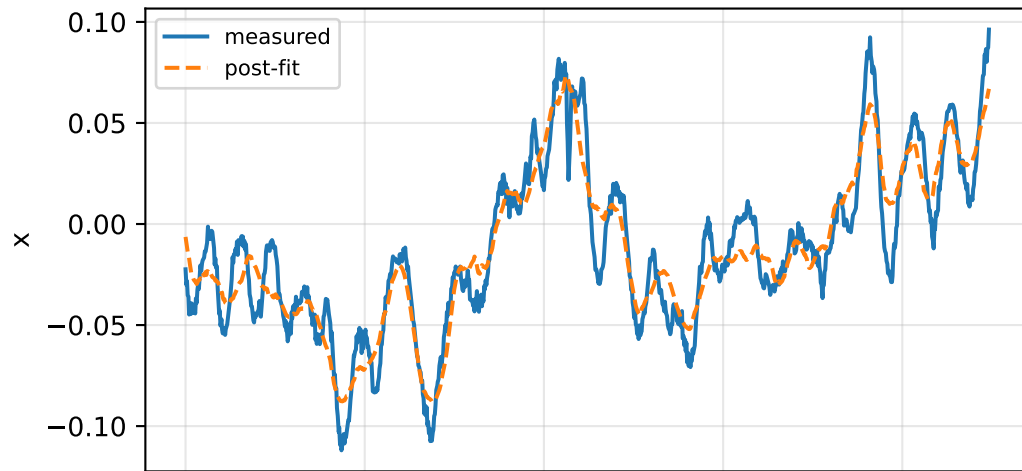


gimbal 4: RMSE=0.006321, max=0.03642 rad

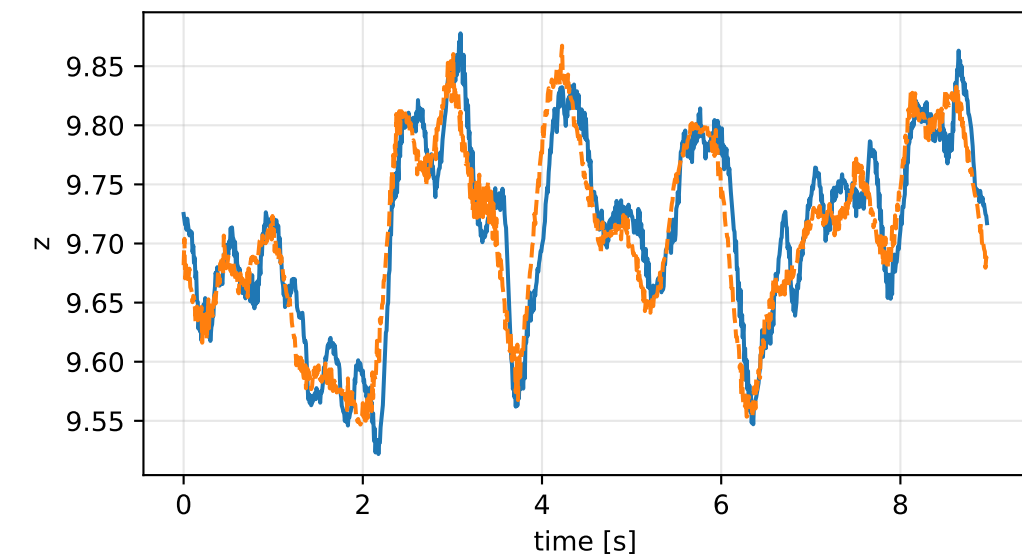
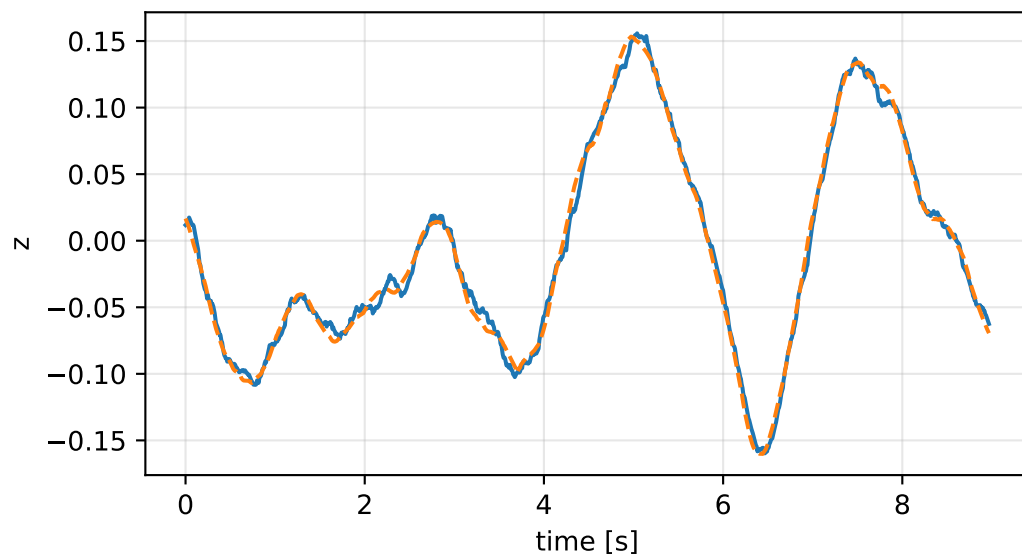
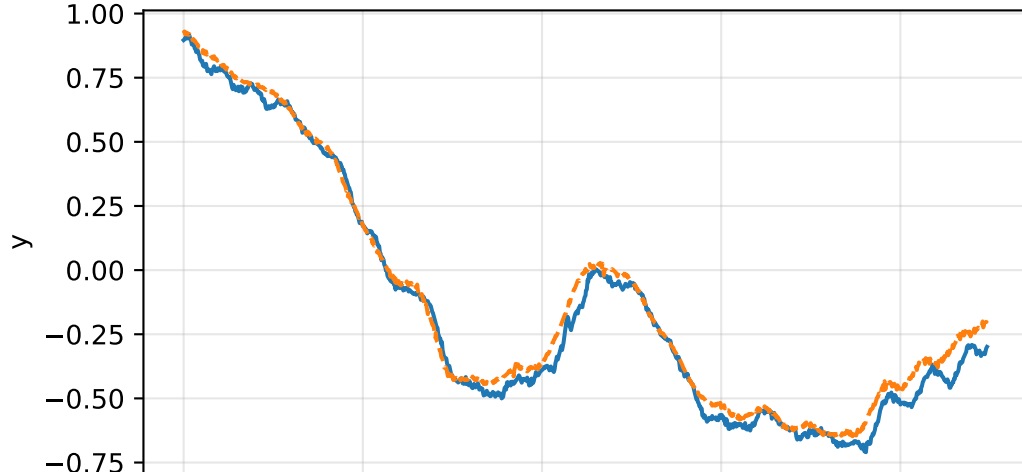
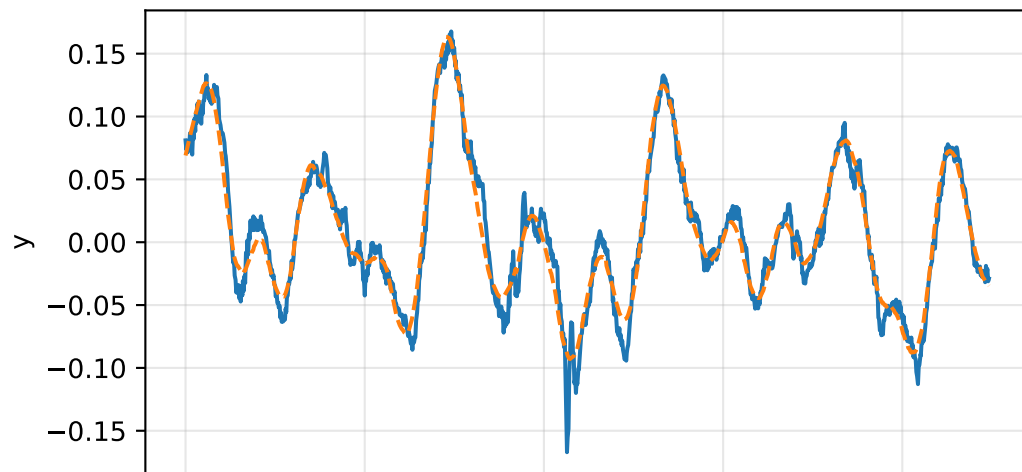
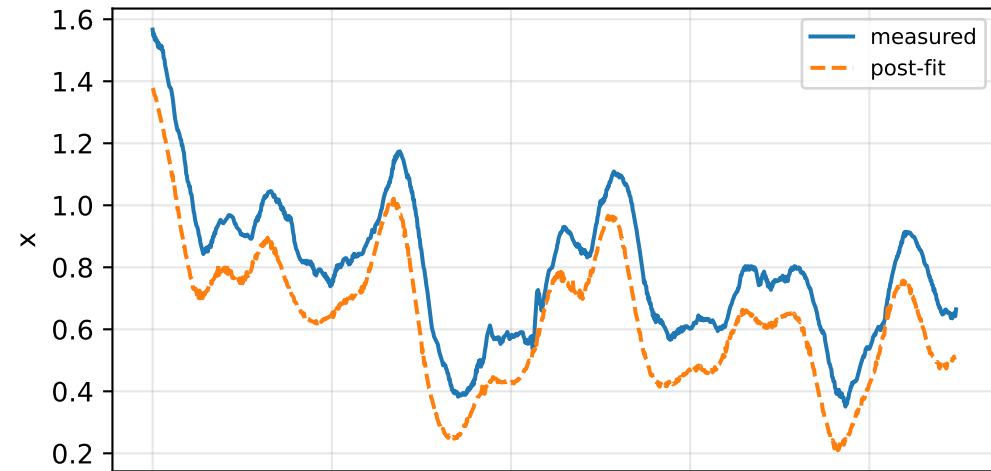


so3_naive_rotation_vector_derivatives: IMU comparison (diagnostic only)

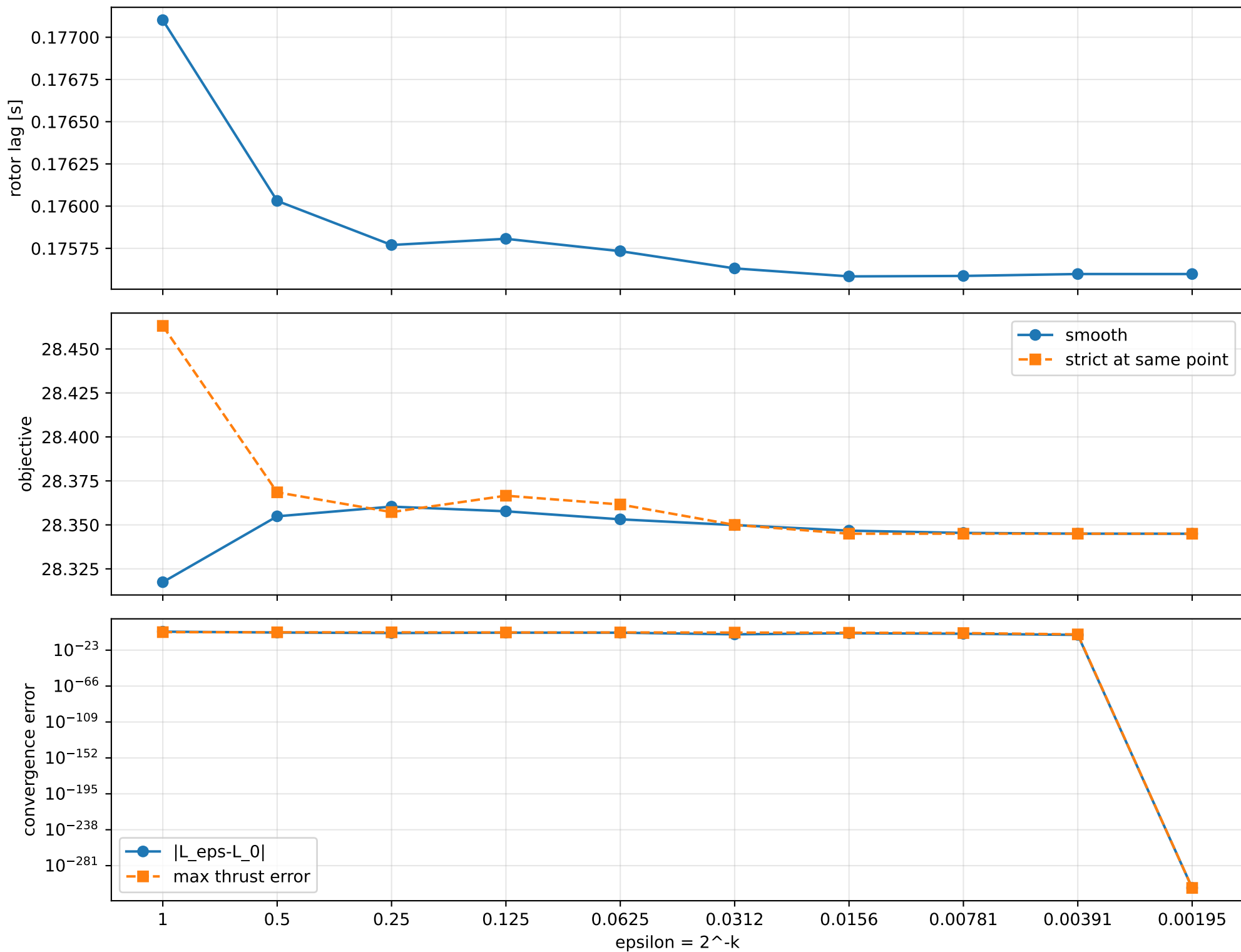
gyro [rad/s]



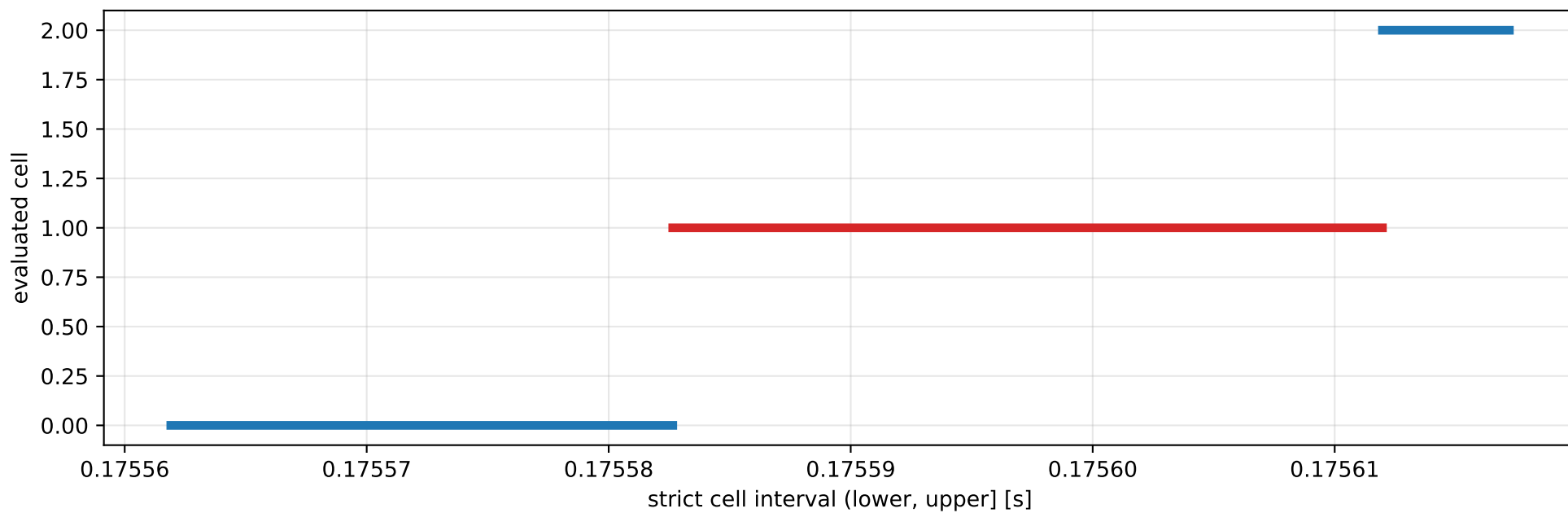
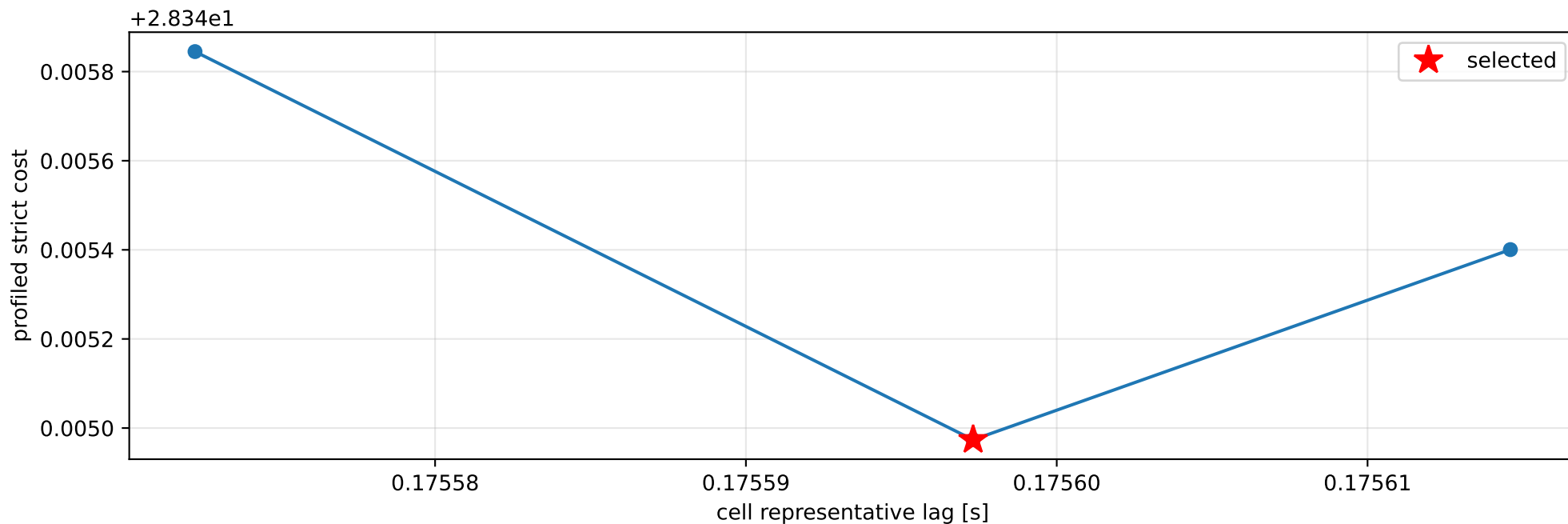
specific force [m/s^2]



so3_naive_rotation_vector_derivatives: smooth-to-strict continuation



so3_naive_rotation_vector_derivatives: exact strict-ZOH lag cells



selected (0.175582647, 0.175611973] s; final smooth 0.175597985 s; data boundary=False

so3_naive_rotation_vector_derivatives: parameter estimate

Case: so3_naive_rotation_vector_derivatives

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 5.95371998

inertia [kg m²]:

```
[[ 1.38094527 -0.0306429  0.10741256]
 [-0.0306429  0.6529745 -0.01825592]
 [ 0.10741256 -0.01825592  0.76745801]]
```

CoG body [m]: [-0.01982347 -0.05512847 -0.00502933]

force effectiveness: [2.64660846 2.58985675 1.36643893 1.65920012]

scale-free J/m [m²]:

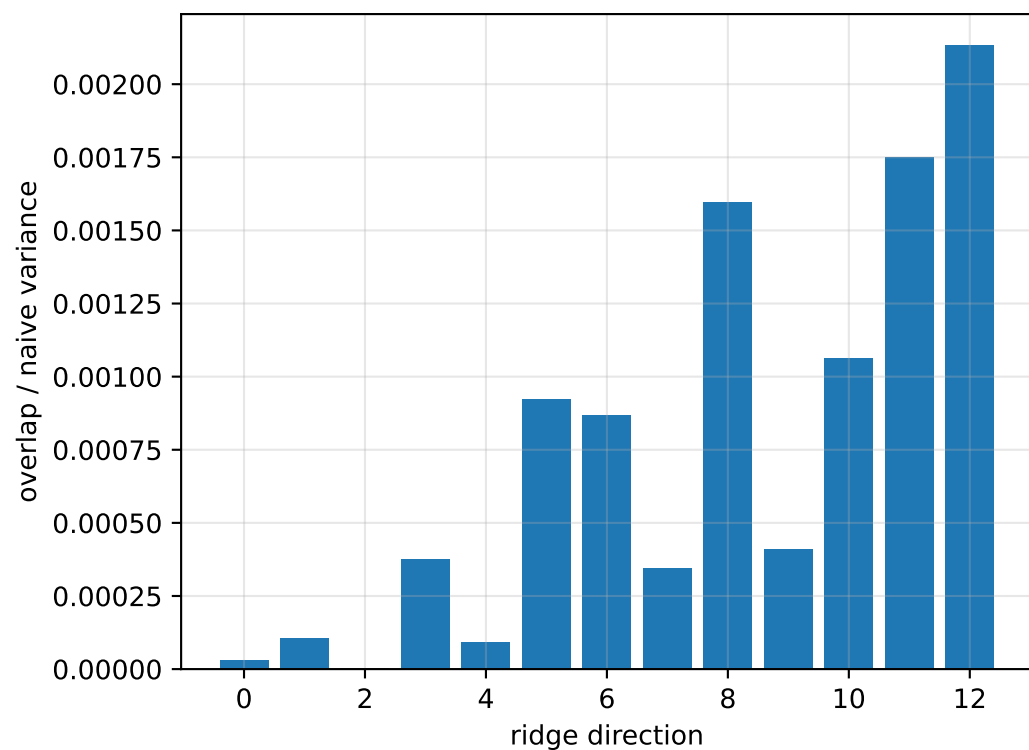
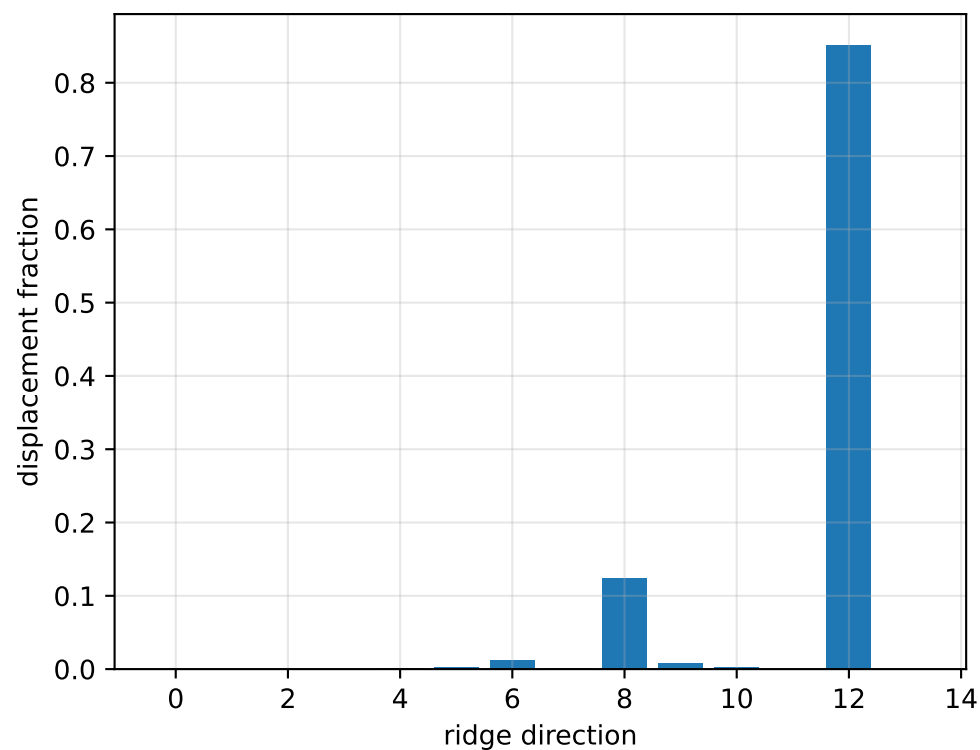
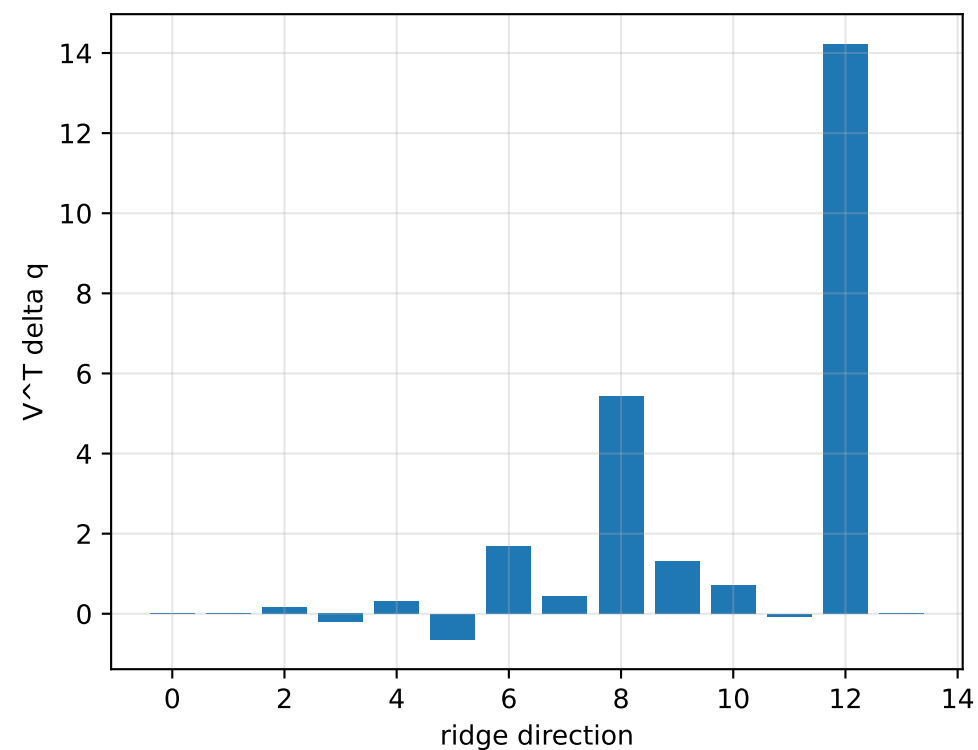
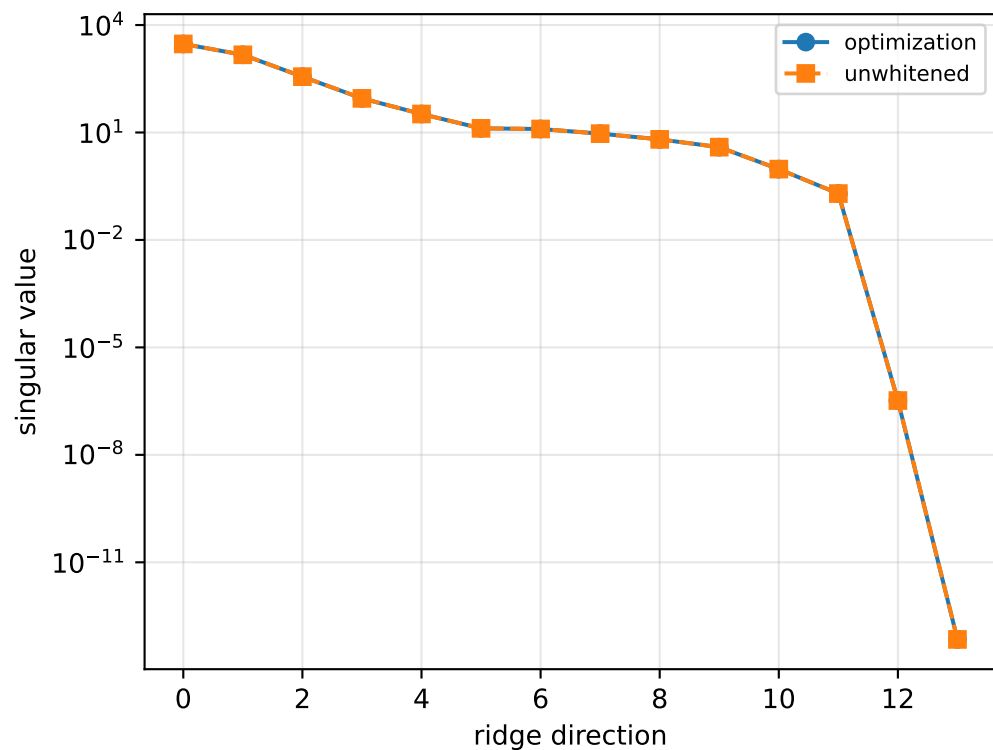
```
[[ 0.23194663 -0.00514685  0.01804125]
 [-0.00514685  0.10967504 -0.00306631]
 [ 0.01804125 -0.00306631  0.12890395]]
```

scale-free f/m: [0.44453022 0.43499808 0.22951011 0.27868293]

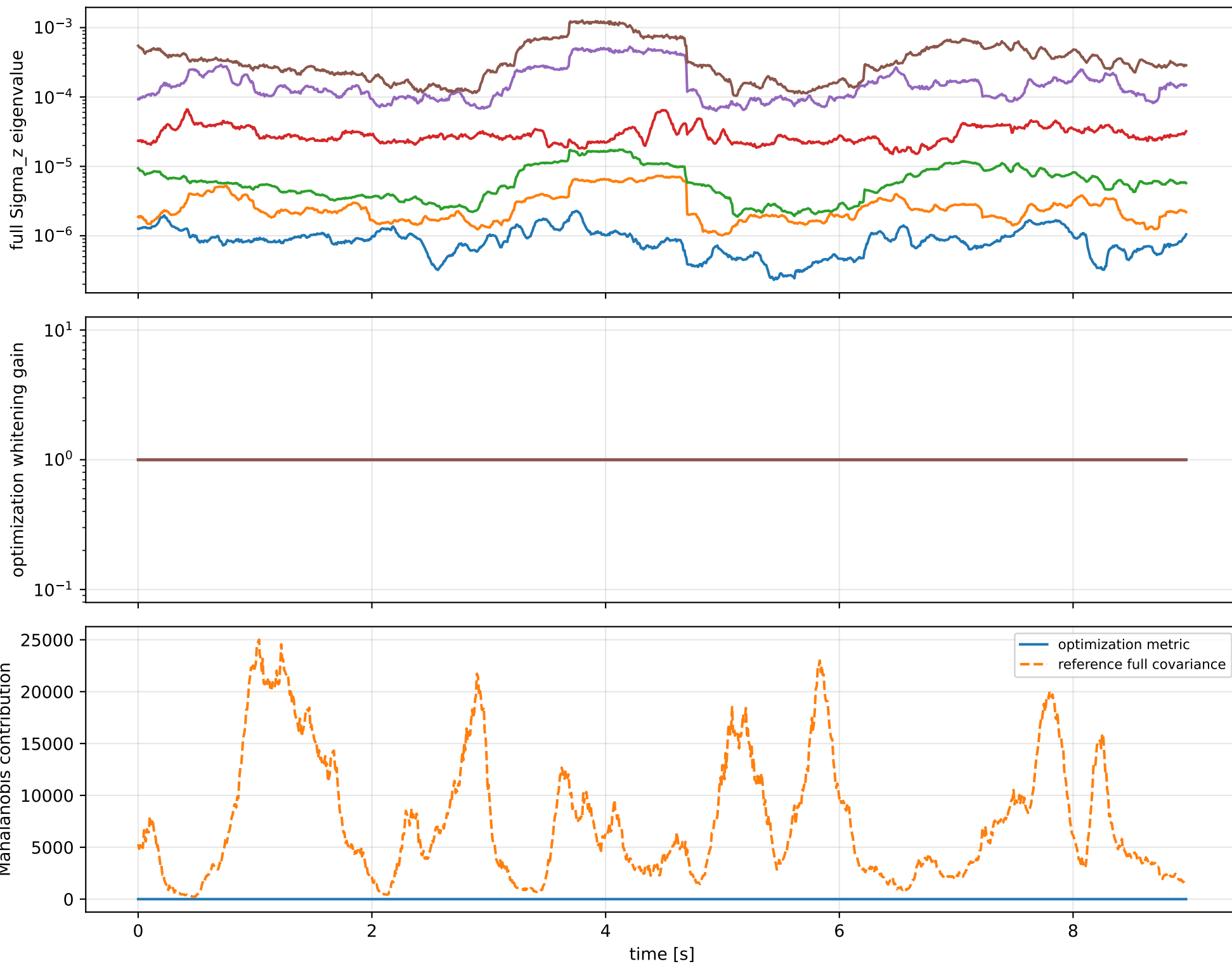
rotor lag cell: (0.175582647, 0.175611973] s

representative: 0.17559731 s

so3_naive_rotation_vector_derivatives: ridge spectrum (rank 13, nullity 1, $||\mathbf{v}||=7.98\text{e-}14$)



so3_naive_rotation_vector_derivatives: optimization vs reference covariance



so3_naive_rotation_vector_derivatives: wrench / closure (force max 1.69e-14, torque max 1.39e-16)

